1

2

$$\left[-\frac{d^2}{dx^2}+V(x)\right]\Psi = E\Psi \qquad \text{where} \qquad V(x)=\frac{1}{4}\omega_1^2 x^2. \tag{1}$$

$$V^l(x) = V(x) + \lambda_0 x, \tag{2}$$

$$\tilde{V}^l(\tilde{x}) = \frac{1}{4}\omega_1^2\tilde{x}^2 - \frac{\lambda_0^2}{\omega_1^2} \quad \text{where} \quad \tilde{x} = x + \frac{2\lambda_0}{\omega_1^2}, \tag{3}$$

$$E \to \tilde{E}_n^l(\omega_1) = \left(n+\frac{1}{2}\right)\omega_1 + \frac{\gamma_0^2}{\omega_1^2}; \quad n = 0,1,2... \tag{4}$$

$$\tilde{\psi}_n^l(\tilde{x}) = \psi_n(\tilde{x}), \quad \text{Here,} \quad \tilde{\omega_1} = \omega_1. \tag{5}$$

$$\mathcal{PT}\ \tilde{\psi}_n^l(\tilde{x}) = (-1)^n\ \tilde{\psi}_n^l(\tilde{x}) \tag{6}$$

2.1

$$V_{RE,m}(x) = V(x) + V_{rat,m}(x), \quad -\infty < x < \infty \tag{7}$$

$$V_{rat,m}(x) = -2\left[\frac{\mathcal{H}_m''\left(\sqrt{\frac{\omega_1}{2}}x\right)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}x\right)} - \left[\frac{\mathcal{H}_m'\left(\sqrt{\frac{\omega_1}{2}}x\right)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}x\right)}\right]^2 + \frac{\omega_1}{2}\right] \tag{8}$$

$$\begin{aligned} \psi_{RE,m,0}(x) &= \frac{\zeta(\omega_1,x)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}x\right)} \\ \text{and} \quad \psi_{RE,m,n+1}(x) &= \frac{\zeta(\omega_1,x)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}x\right)}\hat{H}_{m,n+1}\left(\sqrt{\frac{\omega_1}{2}}x\right), \end{aligned} \tag{9}$$

$$\hat{H}_{m,n+1}(x) = \mathcal{H}_m(x)H_{n+1}(x) + H_n(x)\frac{d}{dx}\mathcal{H}_m(x), \tag{10}$$

$$E_{RE,m,n+1}(\omega_1) = (n+m+1)\,\omega_1 \qquad \text{with} \quad E_{RE,m,0}(\omega_1) = 0. \tag{11}$$

$$\tilde{V}_{RE,m}^l(\tilde{x}) = \tilde{V}^l(\tilde{x}) + V_{rat,m}(\tilde{x}), \quad -\infty < x < \infty \tag{12}$$

$$\begin{aligned} \tilde{\psi}_{RE,m,0}^l(\tilde{x}) &= \frac{\zeta(\omega_1,\tilde{x})}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}\tilde{x}\right)} \\ \text{and} \quad \tilde{\psi}_{RE,m,n+1}^l(\tilde{x}) &= \frac{\zeta(\omega_1,\tilde{x})}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}\tilde{x}\right)}\hat{H}_{m,n+1}\left(\sqrt{\frac{\omega_1}{2}}\tilde{x}\right). \end{aligned} \tag{13}$$

$$\tilde{E}^l_{RE,m,n+1}(\omega_1) = (n+m+1)\,\omega_1 \qquad \text{with} \quad \tilde{E}^l_{RE,m,0}(\omega_1) = 0. \tag{14}$$

$$\begin{aligned} \mathcal{PT}\,\tilde{\psi}^l_{RE,m,0}(\tilde{x}) &= \tilde{\psi}^l_{RE,m,0}(\tilde{x}) \\ \text{and} \quad \mathcal{PT}\,\tilde{\psi}^l_{RE,m,n+1}(\tilde{x}) &= (-1)^{n+1}\,\tilde{\psi}^l_{RE,m,n+1}(\tilde{x}) \end{aligned} \tag{15}$$

3

$$V(x,y) = \frac{1}{4}\omega_1^2 x^2 + \frac{1}{4}\omega_2^2 y^2 + \frac{\lambda}{2}xy. \tag{16}$$

$$\begin{matrix} x = a\tilde{x} + b\tilde{y}, \\ y = -b\tilde{x} + a\tilde{y}, \end{matrix} \begin{cases} \tilde{x} = ax - by, \\ \tilde{y} = bx + ay, \end{cases} \tag{17}$$

$$\begin{aligned} \tilde{\omega}_1 &= \sqrt{\frac{1}{2}\left(\omega_1^2 + \omega_2^2 - \sqrt{4\lambda^2 + (\omega_1^2 - \omega_2^2)^2}\right)} \\ \text{and} \quad \tilde{\omega}_2 &= \sqrt{\frac{1}{2}\left(\omega_1^2 + \omega_2^2 + \sqrt{4\lambda^2 + (\omega_1^2 - \omega_2^2)^2}\right)} \end{aligned} \tag{18}$$

$$\tilde{V}(\tilde{x},\tilde{y}) = \tilde{V}(\tilde{x}) + \tilde{V}(\tilde{y}). \tag{19}$$

$$\tilde{\psi}_{n_1,n_2}(\tilde{x},\tilde{y}) \propto \tilde{\psi}_{n_1}(\tilde{x})\,\tilde{\psi}_{n_2}(\tilde{y}), \quad n_1, n_2 = 0, 1, 2, \cdots, \tag{20}$$

$$\text{and} \quad \tilde{E}_{n_1,n_2}(\tilde{\omega}_1,\tilde{\omega}_2) = \left(n_1 + \frac{1}{2}\right)\tilde{\omega}_1 + \left(n_2 + \frac{1}{2}\right)\tilde{\omega}_2, \tag{21}$$

$$\sqrt{4\lambda^2 + (\omega_1^2 - \omega_2^2)^2} \leq \omega_1^2 + \omega_2^2, \tag{22}$$

$$\begin{aligned} &\sqrt{-4\gamma^2 + (\omega_1^2 - \omega_2^2)^2} \leq \omega_1^2 + \omega_2^2, \\ \text{and} \quad &-4\gamma^2 + (\omega_1^2 - \omega_2^2)^2 > 0. \end{aligned} \tag{23}$$

$$\eta = \begin{pmatrix} -k & -\sqrt{1-k^2} \\ \sqrt{1-k^2} & -k \end{pmatrix} \tag{24}$$

$$V^\dagger = \eta V \eta^{-1}. \tag{25}$$

3.1

$$\tilde{V}_{RE,m_1,m_2}(\tilde{x},\tilde{y}) = \tilde{V}_{RE,m_1}(\tilde{x}) + \tilde{V}_{RE,m_2}(\tilde{y}), \tag{26}$$

$$\tilde{\psi}_{RE,m_1,m_2,0,0}(\tilde{x},\tilde{y}) \propto \tilde{\psi}_{RE,m_1,0}(\tilde{x})\ \tilde{\psi}_{RE,m_2,0}(\tilde{y}) \tag{27}$$

$$\tilde{\psi}_{RE,m_1,m_2,n_1+1,n_2+1}(x,y) \propto \tilde{\psi}_{RE,m_1,n_1+1}(\tilde{x})\ \tilde{\psi}_{RE,m_2,n_2+1}(\tilde{y}) \tag{28}$$

$$\text{and}\quad \tilde{E}_{RE,m_1,m_2,n_1+1,n_2+1}(\tilde{\omega}_1,\tilde{\omega}_2) = \tilde{E}_{RE,m_1,n_1+1}(\tilde{\omega}_1) + \tilde{E}_{RE,m_2,n_2+1}(\tilde{\omega}_2) \tag{29}$$

$$\text{with}\quad \tilde{E}_{RE,m_1,m_2,0,0}(\tilde{\omega}_1,\tilde{\omega}_2) = 0, \quad n_1,n_2 = 0,1,2......,$$

3.2

$$\frac{\tilde{\omega}_1}{\tilde{\omega}_2} = \frac{\tilde{r}}{\tilde{s}}, \tag{30}$$

$$\tilde{V}(\tilde{x},\tilde{y}) = \frac{1}{4}\left(\tilde{\omega}_1^2\tilde{x}^2 + \tilde{\omega}_2^2\tilde{y}^2\right). \tag{31}$$

$$\lambda(r,s,\tilde{r},\tilde{s},\omega_1,\omega_2) = \frac{\sqrt{(\tilde{r}^4+\tilde{s}^4) - \left(\frac{\tilde{r}\tilde{s}}{rs}\right)^2(r^4+s^4)}}{\tilde{r}^2+\tilde{s}^2}\,\omega_1\omega_2 \tag{32}$$

$$\tilde{E}_{n_1,n_2}(\tilde{r},\tilde{s},\tilde{\omega}_2) = \left(\tilde{r}n_1 + \tilde{s}n_2 + \frac{\tilde{r}+\tilde{s}}{2}\right)\frac{\tilde{\omega}_2}{\tilde{s}}. \tag{33}$$

3.2.1

$$\tilde{x} = \frac{1}{\sqrt{2}}(x+y), \qquad \tilde{y} = \frac{1}{\sqrt{2}}(y-x), \qquad \begin{cases} \tilde{\omega}_1 = \sqrt{\omega^2-\lambda}, \\ \tilde{\omega}_2 = \sqrt{\omega^2+\lambda}. \end{cases} \tag{34}$$

$$\lambda(\tilde{r},\tilde{s},\omega) = \frac{|\tilde{r}^2-\tilde{s}^2|}{\tilde{r}^2+\tilde{s}^2}\,\omega^2. \tag{35}$$

$$E_{\lambda=0} = (n_1+n_2+1)\,\omega \tag{36}$$

$$V(x,y) = \frac{1}{4}\left(\frac{1}{2}x^2 + \frac{9}{2}y^2\right) + \frac{\sqrt{7}}{4}xy. \tag{37}$$

3.2.2

$$V(x,y) = \frac{1}{4}\left(x^2+9y^2\right) + i\frac{\sqrt{7}}{2}xy. \tag{38}$$

4

4.1

$$V^{lq}(x,y,z) = \frac{1}{4}\left(\omega_x^2 x^2 + \omega_y^2 y^2 + \omega_z^2 z^2\right) + \lambda_0 z + \frac{\lambda}{2}xy. \tag{39}$$

$$V^{lq}(x,y,z) = V(x,y) + V^l(z), \tag{40}$$

$$\tilde{V}^{lq}(\tilde{x},\tilde{y},\tilde{z}) = \tilde{V}(\tilde{x},\tilde{y}) + \tilde{V}^l(\tilde{z}). \tag{41}$$

$$\tilde{\psi}^{lq}_{n_1,n_2,n_3}(\tilde{x},\tilde{y},\tilde{z}) \propto \tilde{\psi}_{n_1,n_2}(\tilde{x},\tilde{y})\tilde{\psi}^l_{n_3}(\tilde{z}) \tag{42}$$

$$\tilde{E}^{lq}_{n_1,n_2,n_3}(\tilde{\omega}_1,\tilde{\omega}_2,\tilde{\omega}_3) = \tilde{E}_{n_1,n_2}(\tilde{\omega}_1,\tilde{\omega}_2) + \tilde{E}_{n_3}(\tilde{\omega}_3) \tag{43}$$

$$\mathcal{P}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}. \tag{44}$$

$$\mathcal{P}_2 T\ \tilde{\psi}^{lq}_{n_1,n_2,n_3}(\tilde{x},\tilde{y},\tilde{z}) = (-1)^{n_3}\tilde{\psi}^{lq}_{n_1,n_2,n_3}(\tilde{x},\tilde{y},\tilde{z}).$$

$$\mathcal{P}_1 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \qquad \mathcal{P}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{45}$$

$$\mathcal{P}_1 T\ \tilde{\psi}^{lq}_{n_1,n_2,n_3}(\tilde{x},\tilde{y},\tilde{z}) = \mathcal{P}_3 T \psi^{lq}_{n_1,n_2,n_3}(\tilde{x},\tilde{y},\tilde{z}) = (-1)^{n_1+n_2}\tilde{\psi}^{lq}_{n_1,n_2,n_3}(\tilde{x},\tilde{y},\tilde{z}).$$

$$\mathcal{P}_4 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \tag{46}$$

$$\mathcal{P}_4 T\ \tilde{\psi}^{lq}_{n_1,n_2,n_3}(\tilde{x},\tilde{y},\tilde{z}) = (-1)^{n_1+n_2+n_3}\tilde{\psi}^{lq}_{n_1,n_2,n_3}(\tilde{x},\tilde{y},\tilde{z}).$$

4.1.1

$$\tilde{V}^{lq}_{RE,m_1,m_2,m_3}(\tilde{x},\tilde{y},\tilde{z}) = \tilde{V}_{RE,m_1,m_2}(\tilde{x},\tilde{y}) + \tilde{V}^l_{RE,m_3}(\tilde{z}) \tag{47}$$

$$\tilde{\psi}^{lq}_{RE,m_1,m_2,m_3,0,0,0}(\tilde{x},\tilde{y},\tilde{z}) \propto \tilde{\psi}_{RE,m_1,m_2,0,0}(\tilde{x},\tilde{y})\ \tilde{\psi}^l_{RE,m_3,0}(\tilde{z}), \tag{48}$$

$$\tilde{\psi}^{lq}_{RE,m_1,m_2,m_3,n_1+1,n_2+1,n_3+1}(\tilde{x},\tilde{y},\tilde{z}) \propto \tilde{\psi}_{RE,m_1,m_2,n_1+1,n_2+1}(\tilde{x},\tilde{y})\ \tilde{\psi}^l_{RE,m_3,n_3+1}(\tilde{z}) \tag{49}$$

$$\tilde{E}^{lq}_{RE,m_1,m_2,m_3,n_1+1,n_2+1,n_3+1}(\tilde{\omega}_1,\tilde{\omega}_2,\tilde{\omega}_3) = \tilde{E}_{RE,m_1,m_2,n_1+1,n_2+1}(\tilde{\omega}_1,\tilde{\omega}_2) + \tilde{E}^l_{RE,m_3,n_3+1}(\tilde{\omega}_3) \tag{50}$$

$$\tilde{E}^{lq}_{RE,m_1,m_2,m_3,0,0,0}(\tilde{\omega}_1,\tilde{\omega}_2,\tilde{\omega}_3) = \tilde{E}_{RE,m_1,m_2,0,0}(\tilde{\omega}_1,\tilde{\omega}_2) + \tilde{E}^l_{RE,m_3,0}(\tilde{\omega}_3)$$

4.2

$$V(x,y,z)=\frac{1}{4}\left(\omega_x^2x^2+\omega_y^2y^2+\omega_z^2z^2+2\lambda_1xy+2\lambda_2yz+2\lambda_3zx\right). \tag{51}$$

$$V^q(x,z,y)=\frac{1}{4}\left(\,\omega^2(x^2+y^2)+\omega_z^2z^2+2\lambda_1xy+2\lambda_2yz+2\lambda_3zx\right). \tag{52}$$

4.2.1

$$V^{q_1}(x,y,z)=\frac{1}{4}\left(\omega^2(x^2+y^2)+\omega_z^2z^2+2\lambda_2yz+2\lambda_3zx\right) \tag{53}$$

$$\begin{matrix} x=ad\tilde{y}+bd\tilde{z}-c\tilde{x}, \\ y=d\,\tilde{x}+ac\tilde{y}+bc\tilde{z} \\ z=a\tilde{z}-b\tilde{y}, \end{matrix} \begin{cases} \tilde{x}=-cx+d\,y, \\ \tilde{y}=ad\,x+acy-bz, \\ \tilde{z}=bd\,x+bcy+az. \end{cases} \tag{54}$$

$$\tilde{V}^q(\tilde{x},\tilde{y},\tilde{z})=\frac{1}{4}\left(\tilde{\omega}_1^2\tilde{x}^2+\tilde{\omega}_2^2\tilde{y}^2+\tilde{\omega}_3^2\tilde{z}^2\right) \tag{55}$$

$$\begin{aligned} \tilde{\omega}_1&=\omega, \\ \tilde{\omega}_2&=\sqrt{\frac{1}{2}\left(\omega^2+\omega_3^2-\sqrt{4(\lambda_2^2+\lambda_3^2)+\left(\omega^2-\omega_3^2\right)^2}\right)}, \\ \tilde{\omega}_3&=\sqrt{\frac{1}{2}\left(\omega^2+\omega_3^2+\sqrt{4(\lambda_2^2+\lambda_3^2)+\left(\omega^2-\omega_3^2\right)^2}\right)}. \end{aligned} \tag{56}$$

$$\tilde{V}^{q_1}_{RE,m_1,m_2,m_3}(\tilde{x},\tilde{y},\tilde{z})=\tilde{V}_{RE,m_1}(\tilde{x})+\tilde{V}_{RE,m_2}(\tilde{y})+\tilde{V}_{RE,m_3}(\tilde{z}) \tag{57}$$

$$\tilde{\psi}^{q_1}_{RE,m_1,m_2,m_3,0,0,0}(\tilde{x},\tilde{y},\tilde{z})\propto\tilde{\psi}_{RE,m_1,0}(\tilde{x})\;\tilde{\psi}_{RE,m_2,0}(\tilde{y})\;\tilde{\psi}_{RE,m_3,0}(\tilde{z})$$

$$\tilde{\psi}^{q_1}_{RE,m_1,m_2,m_3,n_1+1,n_2+1,n_3+1}(\tilde{x},\tilde{y},\tilde{z})\propto\tilde{\psi}_{RE,m_1,n_1+1}(\tilde{x})\;\tilde{\psi}_{RE,m_2,n_2+1}(\tilde{y})\;\tilde{\psi}_{RE,m_3,n_3+1}(\tilde{z}) \tag{58}$$

$$\begin{aligned} \tilde{E}^{q_1}_{RE,m_1,m_2,m_3,n_1+1,n_2+1,n_3+1}(\tilde{\omega}_1,\tilde{\omega}_2,\tilde{\omega}_3)&=\tilde{E}_{RE,m_1,n_1+1}\left(\tilde{\omega}_1\right) \\ &+\tilde{E}_{RE,m_2,n_2+1}\left(\tilde{\omega}_2\right)+\tilde{E}_{RE,m_3,n_3+1}\left(\tilde{\omega}_3\right) \end{aligned} \tag{59}$$

$$\tilde{E}^{q_1}_{RE,m_1,m_2,m_3,0,0,0}(\tilde{\omega}_1,\tilde{\omega}_2,\tilde{\omega}_3)=0 \tag{60}$$

$$0\le\lambda_2^2+\lambda_3^2\le\omega^2\omega_3^2 \tag{61}$$

$$\sqrt{\lambda_2^2+\lambda_3^2}=\frac{\sqrt{(\tilde{u}^4+\tilde{v}^4)-\left(\frac{\tilde{u}\tilde{v}}{uv}\right)^2(u^4+v^4)}}{\tilde{u}^2+\tilde{v}^2}\;\omega\omega_3 \tag{62}$$

$$\sqrt{\lambda_2^2+\lambda_3^2}=\frac{\sqrt{2(\tilde{u}^2-\tilde{v}^2)^2-(\tilde{u}\tilde{v})^2}}{\tilde{u}^2+\tilde{v}^2}\;\omega^2 \tag{63}$$

4.2.2

$$V^{q_2}(x,y,z) = \frac{1}{4}\left(\omega^2(x^2+y^2) + \omega_z^2 z^2 + 2\lambda_1 xy + 2\lambda(yz+zx)\right). \tag{64}$$

$$\begin{aligned} x &= \frac{a}{\sqrt{2}}\tilde{y} + \frac{b}{\sqrt{2}}\tilde{z} - \frac{\tilde{x}}{\sqrt{2}}, \\ y &= \frac{a}{\sqrt{2}}\tilde{y} + \frac{b}{\sqrt{2}}\tilde{z} + \frac{\tilde{x}}{\sqrt{2}}, \\ z &= a\tilde{z} - b\tilde{y}, \end{aligned} \quad \begin{cases} \tilde{x} = \dfrac{y}{\sqrt{2}} - \dfrac{x}{\sqrt{2}}, \\ \tilde{y} = \dfrac{a}{\sqrt{2}}x + \dfrac{a}{\sqrt{2}}y - bz, \\ \tilde{z} = \dfrac{b}{\sqrt{2}}x + \dfrac{b}{\sqrt{2}}y + az. \end{cases} \tag{65}$$

$$\begin{aligned} \tilde{\omega}_1 &= \sqrt{\omega^2 - \lambda_1}, \\ \tilde{\omega}_2 &= \sqrt{\frac{1}{2}\left(\omega^2 + \omega_3^2 + \lambda_1 - \sqrt{8\lambda^2 + (\omega^2 - \omega_3^2 + \lambda_1)^2}\right)}, \\ \tilde{\omega}_3 &= \sqrt{\frac{1}{2}\left(\omega^2 + \omega_3^2 + \lambda_1 + \sqrt{8\lambda^2 + (\omega^2 - \omega_3^2 + \lambda_1)^2}\right)}. \end{aligned} \tag{66}$$

$$\begin{gathered} -\omega^2 \leq \lambda_1 \leq \omega^2 \\ \text{and} \quad 0 \leq \lambda_1^2 \leq \frac{(\omega^2+\lambda)\omega_z^2}{2}. \end{gathered} \tag{67}$$

$$-\omega^2 \leq \lambda_1 \leq \omega^2 \quad \text{and} \quad 0 \leq \gamma^2 \leq \frac{(\omega^2 - \omega_3^2 + \lambda_1)^2}{8} \tag{68}$$

$$\sqrt{8\lambda^2 + \lambda_1^2} = \frac{|\tilde{v}^2 - \tilde{u}^2|}{\tilde{v}^2 + \tilde{u}^2}\,(2\omega^2 + \lambda_1), \tag{69}$$

$$\lambda = \frac{\sqrt{\left(\frac{|\tilde{v}^2 - \tilde{u}^2|}{\tilde{v}^2 + \tilde{u}^2}(\lambda_1 + 2\omega^2)\right)^2 - \lambda_1^2}}{2\sqrt{2}}. \tag{70}$$

5

$$\hat{H}\,|\psi_n\rangle = E_n\,|\psi_n\rangle\,, \tag{71}$$

$$\hat{H}_0 = U\hat{H}U^{-1}, \tag{72}$$

$$|\psi_n\rangle = U\,|\tilde{\psi}_n\rangle\,. \tag{73}$$

$$\hat{H}_0 = \left(U^{-1}\right)^\dagger \hat{H}^\dagger U^\dagger. \tag{74}$$

$$|\Psi_n\rangle = \left(U^{-1}\right)^\dagger |\tilde{\psi}_n\rangle\,. \tag{75}$$

$$\langle\psi_n|\psi_m\rangle = \langle\tilde{\psi}_n|U^\dagger U|\tilde{\psi}_m\rangle = \langle\tilde{\psi}_n|\eta|\tilde{\psi}_m\rangle = \delta_{nm}. \tag{76}$$

$$\langle\psi_n|\psi_m\rangle = \langle\tilde{\psi}_n|\left(U^{-1}\right)^\dagger U^{-1}|\tilde{\psi}_m\rangle = \delta_{nm}, \tag{77}$$

$$\hat{H}^\dagger = \eta\hat{H}\eta^{-1}. \tag{78}$$

$$[A,B]_\psi = U^{-1}\left[A',B'\right]_{\tilde{\psi}} U. \tag{79}$$

$$[x,p] = iI. \tag{80}$$

$$[H,\mathcal{PT}] = 0 \quad \text{and} \quad \mathcal{PT}\left|\psi\right\rangle = \pm\left|\psi\right\rangle, \tag{81}$$

$$\hat{H} = \chi\hat{H}\chi^{-1}, \tag{82}$$

$$\mathcal{P}_1 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}; \quad \text{and} \quad \mathcal{P}_2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \tag{83}$$

$$\mathcal{P}_3 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}; \quad \text{and} \quad \mathcal{P}_4 = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \tag{84}$$

$$\begin{aligned} \mathcal{P}_1 &= \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \qquad \mathcal{P}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}; \\ \mathcal{P}_3 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \qquad \mathcal{P}_4 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}. \end{aligned} \tag{85}$$